Top 10 medical photography tips for a camera phone

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A picture is worth a thousand words.

My corollary to this statement is that a poorly framed or blurry image significantly detracts from its impactfulness. Plus, it just looks unprofessional. I have had to either retake or Photoshop-edit several photos submitted for blog posts. There have been many amazing photos which I decided not to use because of image quality.

Dr. Jason Thurman, under the mentorship of **Dr. Larry Stack** (both at Vanderbilt University), recently shared his thoughts about medical photography. I approached him because he gave a wonderful SAEM lecture on this. Although his talk focused primarily on optimizing images using a SLR camera (nice review by Dr. Rob Cooney), there are many principles which hold true for camera phones.

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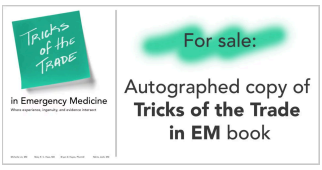

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My point is that most clinicians don't have a SLR camera on shift. What we do have, however, are cameras on our iPhones or Androids. It's not ideal, but it's way better than any crayon-sketch I can do.

Below are some tips to make the best of camera-phone medical photography, which I adopted from Jason's teaching points. Note that if you want to take truly excellent medical photographs, you will need to make the investment for proper camera and lighting equipment.

1. Be sure the image is in focus!

- This seems obvious, but be sure the image of interest is crisp. The image above shows a blurry image because the camera focused on the background. Use the zoom feature of your image preview to double check.
- Hold your camera very still while taking the photo.
- Don't hold the camera too close to the image. Camera phones have poor macro capability. It's better to move a few centimeters away and then later crop/zoom the image.

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2. Manage distracting elements to make the image stand out

- Remove ECG leads, oxygen tubing, jewelry, if irrelevant.
- Brush aside hair.

So many things wrong with this photo that I'm embarrassed to share from my photo collection. Background, offset lighting, distractors, oh my.

3. Control the background

- Your primary image should not be overshadowed by distracting backgrounds.
- Use a white, black, or blue background, ideally covering the entire frame of view.
- Sterile towels are excellent.
- Be careful to get everything out of the frame of view (curtain, bedrail, etc)

4. Frame the image with a reference shot

- Example: If you are taking an image of olecranon bursitis, first shoot the entire arm to show the location of the pathology and orient the viewer, then get a close up of the lesion you wish to show.

5. Manage the lighting

- In the ideal world, an external flash provides the best lighting.
- Get creative with lights to make sure the pathology is illuminated as best as possible. This requires taking several photos with a light source being further or closer to the image to avoid whiting-out the image. Consider using the High Dynamic Range (HDR) setting for iPhones.
- Try to avoid using the camera phone flash.

6. Manage perspective distortion

- If you are shooting the face, take images straight on and perpendicular, do not take images at angles as this causes image distortion and poorly represents what you are trying to show)

Ear foreign body

7. Provide a reference for scale

- You know what the image is, but think from the perspective of the viewer. How big exactly is that abscess?
- Place a ruler or commonly identifiable object (eg. coin, pen) next to the object to give the viewer an idea of size.

8. Be your own worst critic

- NEVER be satisfied with a mediocre image.
- Keep shooting until you get it right!

9. Avoid any patient identifier features.

10. Be sure you have your patient's signed consent.

- I bundle my digital photos with a photo of the patient's signed consent form for tracking purposes

as the last page.

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